General notes on chaos theory

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Table 1 Comparison of proposed enhanced logistic map with seed and existing chaotic maps

Existing maps	Equation	Control parameter range
Seed maps		
Logistic Map (May 1976)	$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$	$r \in [3.54, 4]$
Sine Map (Griffin 2013)	$x_{n+1} = \beta \cdot \sin(\pi \cdot x_n)$	$\beta \in [0.87, 1]$
Chebyshev Map (Liu et al. 2016)	$x_{n+1} = \cos\left(\gamma \cdot \cos^{-1}(x_n)\right)$	$-1 \le x_l \le 1, \gamma \in [2, 6]$
Quadratic Map (Ramadan et al. 2016)	$x_{n+1} = \delta - x_n^2$	$\delta \in 1.5$ to 2
Simple Quadratic Map (Muthu and Murali 2021b)	$x_{n+1} = \epsilon\left(x_n^2 - 5\right)$	$0.53 < \epsilon < 0.61$
Singer Map (Tubishat et al. 2022)	$x_{n+1} = \theta\left(7.86x_n - 23.31x_n^2 + 28.75x_n^3 - 13.302875x_n^4\right)$	$\theta \in 0.9$ to 1.08
Existing Modified Maps		
Generalised Logistic Map (Krishnamoorthi and Murali 2017; Bhatnagar and Jonathan Wu 2012)	$f(x) = \frac{4\mu x(1-x)}{1+4(\mu^2-1)x(1-x)}$	$\mu \in [-0.6795, -0.4324]$
Enhanced Logistic Map (Hua et al. 2019)	$x_{n+1} = \sin\left(\pi \bar{\alpha} x_n (1 - x_n)\right)$	$\bar{\alpha} \in (0, +\infty)$
Squared Sine Logistic Map (Carvalho and Leonel 2016)	$f(x) = rx(1 - x) + \mu \sin^2(2\pi x)$	$r \in [3.5, 3.6]$
Logistic-Sine–Cosine (LSC) map (Hua et al. 2019)	$x_{n+1} = \cos\left(\pi\left(4rx_n(1 - x_n) + (1 - r)\sin(\pi x_n) - 0.5\right)\right)$	$r \in [0, 1]$
Tent-logistic-cosine (TLC) map (Hua et al. 2019)	$x_{n+1} = \begin{cases} \cos(\pi(2rx_n + 4(1-r)x_n(1-x_n) - 0.5)), & x_n < 0.5 \\ \cos(\pi(2r(1-x_n) + 4(1-r)x_n(1-x_n) - 0.5)), & x_n \ge 0.5 \end{cases}$	$r \in [0, 1]$
Logistic-tent system (Zhou et al. Apr. 2014)	$x_{n+1} = \begin{cases} (rx_n(1-x_n) + (4-r)x_n/2) \bmod 1 & x_l < 0.5 \\ (rx_n(1-x_n) + (4-r)(1-x_n)/2) \bmod 1 & x_l \ge 0.5 \end{cases}$	$r \in [0, 1]$
Logistic-sine system (Zhou et al. 2014)	$x_{n+1} = (rx_n(1 - x_n) + (4 - r)\sin(\pi x_n)/4) \bmod 1$	$r \in [0, 1]$
Modified Logistic map (Hanis and Amutha 2019)	$z_{n+1} = \mu_z z_n(1 - z_n)\left(1 - z_n^2\right)$	$3.9 < \mu_z \le 6.27$
Improved modified logistic (Attaullah and Jamal 2020)	$x_{n+1} = \left[\mu x_n(1 - x_n) \times 10^j\right] - \mu x_n(1 - x_n) \times 10^j$	$0 \le \mu \le 4$ $2 \le j \le 8$
Proposed 1D Chaotic Map		
Enhanced logistic map	$x_{n+1} = mod\left(\left(\frac{\pi}{x_n}.r'.(1 - x_n)\right), 1\right)$	$r' \in (0, +\infty)$

M Nazish, M Javid, MT Banday "Enhanced logistic map with infinite chaos and its applicability in lightweight and high-speed pseudo-random bit generation." Cybersecurity (2025)

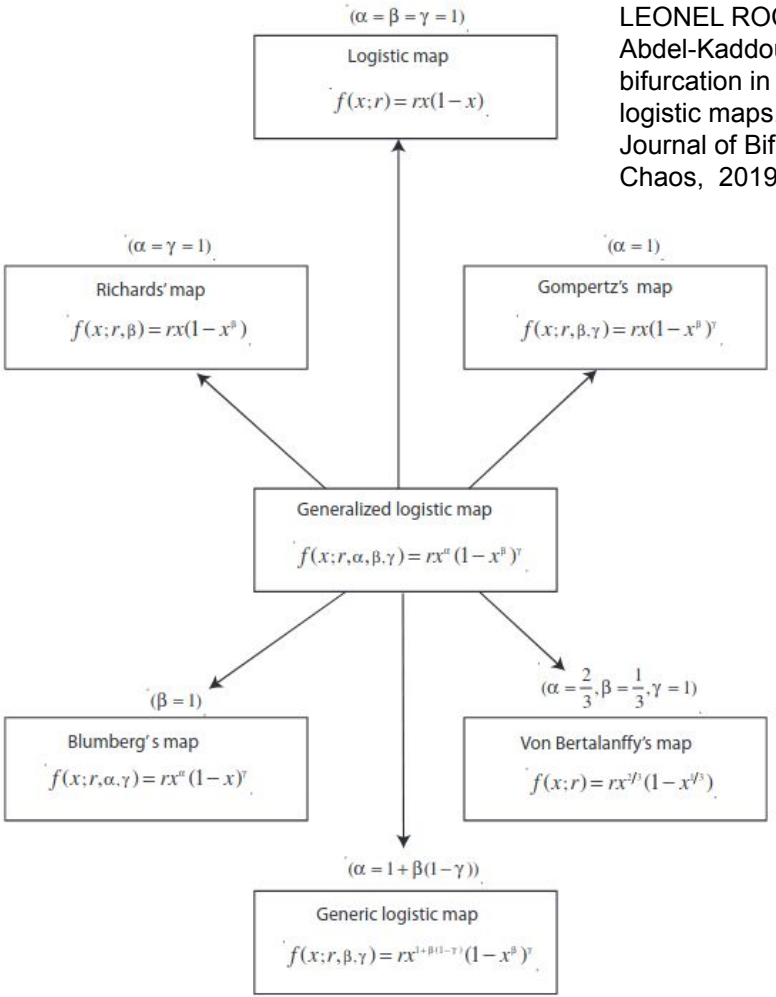


Fig. 1. Scheme of generalized logistic map $f(x;r,\alpha,\beta,\gamma)$ and its derivative maps.

LEONEL ROCHA, J.; TAHA, Abdel-Kaddous. Allee's effect bifurcation in generalized logistic maps. International Journal of Bifurcation and Chaos, 2019.

TABLE 1: Scheme of the generalized logistic map by Rocha et al.

Map	Expression of the sequence
The generalized logistic map	$x_{n+1}=\mu x_n^{\alpha}(1-x_n^{\beta})^{\gamma}$
The generalized discrete logistic map	$x_{n+1}=\mu x_n^{\alpha}(1-x_n^{\alpha})$
Blumberg's map	$x_{n+1}=\mu x_n^{\alpha}(1-x_n)^{\gamma}$
Richards' map	$x_{n+1}=\mu x_n(1-x_n^{\beta})$
Gompertz's map	$x_{n+1}=\mu x_n(1-x_n^{\beta})^{\gamma}$
Generic logistic map	$x_{n+1}=\mu x_n^{1+\beta(1-\gamma)}(1-x_n^{\beta})^{\gamma}$

ZHANG, Li; YU, Wenhui; GAO, Shilong. Chaos of Exponential Logistic Map. Discrete Dynamics in Nature and Society, 2025